



MomentMamba: Efficient collaborative state space modeling for joint video moment retrieval and highlight detection

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ABSTRACT

Joint Video Moment Retrieval (MR) and Highlight Detection (HD) are essential for understanding the ever-growing volume of video content. Existing Transformer-based methods often struggle with misaligned features in complex video-text scenarios and quadratic computational costs. To overcome these bottlenecks, we propose MomentMamba, a unified and efficient framework built upon Selective State Space Models (Mamba). First, we design a multimodal alignment module that integrates Optimal Transport (OT)-based local alignment with a global contrastive strategy, enabling precise and robust semantic alignment between video and text at both token and modality levels. Second, we propose a novel Mamba-based highly-efficient architecture that comprises a Collaborative Mamba module to efficiently capture both modality-specific information and cross-modal interactions, and a Mamba decoder that enables joint MR and HD prediction, which effectively avoids high computational costs. Extensive experiments on QVHighlights, TVSum, and Charades-STA benchmarks demonstrate that MomentMamba consistently outperforms state-of-the-art methods in both MR and HD tasks, while at the same time reducing the theoretical complexity from quadratic $\mathcal{O}(L^2 \cdot d)$ to linear $\mathcal{O}(L \cdot d)$ and achieving a 9% reduction in the average training time per iteration.

1. Introduction

The exponential growth of Web video has made video Moment Retrieval (MR) and Highlight Detection (HD) critical for efficient content understanding. As Fig. 1 shows, MR aims to localize specific temporal boundaries within a video in response to a given textual query, while HD focuses on assigning saliency scores to identify the most significant clips. While early approaches relied on handcrafted features [1,2], recent advancements utilize Transformer-based architectures to model complex video-text interactions [3,4]. These approaches typically employ CLIP-type contrastive learning to align multimodal features in a shared embedding space, followed by attention mechanisms for fusion and decoding.

However, state-of-the-art approaches still face significant limitations in both alignment precision and computational efficiency. Existing methods [5–7] mainly rely on either contrastive objectives for global semantic alignment or attention-based interaction for cross-modal correlation. While contrastive learning effectively enforces the consistency between video and text representations, it often fails to establish fine-grained correspondence between specific query keywords and tempo-

rally localized video clips. Meanwhile, attention-based fusion captures cross-modal dependencies through soft similarity modeling, which lacks explicit token-to-clip matching and may introduce semantic ambiguity. Furthermore, standard Transformer architectures suffer from quadratic computational complexity regarding the number of clips of input video. This high resource consumption hinders the efficient processing of long videos and limits the scalability of temporal modeling. Therefore, advancing MR and HD requires a solution that simultaneously ensures precise multimodal alignment and efficient, scalable video sequence modeling.

To address these challenges, we propose **MomentMamba**, a novel framework designed for precise multimodal alignment and efficient long-sequence modeling. Our approach integrates two key technical innovations: **Optimal Transport (OT) [8] based local-global alignment** and a **highly-efficient architecture based on Selective State Space Models (Mamba) [9]**.

First, to achieve precise alignment between textual and visual features, we introduce a local-global alignment module. This module combines a global alignment with an OT-based local alignment. The global alignment enforces global consistency of paired videos and queries at

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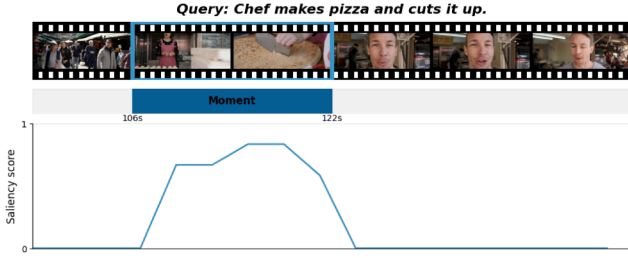


Fig. 1. A visual depiction of video moment retrieval (MR) and highlight detection (HD) tasks. Given an input query “Chef makes pizza and cuts it up.”, the tasks retrieve the most semantically relevant video segments and compute fine-grained saliency scores across time.

a unified shared space through a contrastive objective. The OT mechanism optimizes the transport cost between video clips and text tokens, thereby explicitly capturing token-clip similarity and bridging the representation gap between textual and visual features at the token level. This motivates us to design an OT-based local alignment mechanism to capture clip-token-level semantic relevance.

Second, to overcome the quadratic computational bottleneck inherent in Transformer-based long-sequence modeling, we employ a Mamba backbone, a fully recurrent model with linear computational complexity in sequence length, for the first time of the application in the joint video MR and HD tasks. Based on Mamba backbone [9–13], we propose a collaborative module for cross-modal interaction alongside a decoder for joint temporal reasoning. The collaborative module processes visual and textual features through two parallel Mamba backbones. By sharing core state transition matrices between two parallel processing streams, the collaborative module enables dynamically cross-modal information exchange. Furthermore, we adopt a Mamba backbone for decoding semantically aligned joint features of modalities, which effectively reduces computational complexity from quadratic to linear. This architectural innovation effectively filters out redundant information while highlighting salient clips, concurrently avoiding the heavy computational cost typically associated with traditional attention-heavy models.

Our main contributions are summarized as follows:

- First, we propose MomentMamba, a unified framework that leverages a local-global alignment to enhance both token-clip relevance by an OT-based local alignment and video-query semantic consistency via a contrastive strategy.
- Second, we design a collaborative Mamba-based interaction and decoding architecture, where shared state transition matrices enable cross-modal information exchange while preserving modality-specific dynamics. This design reduces the dominant temporal modeling complexity from quadratic to linear.
- Third, extensive experiments on benchmark datasets demonstrate that MomentMamba achieves state-of-the-art performance, validating its superior effectiveness and high efficiency.

2. Related work

2.1. Multimodal alignment in MR and HD

Multimodal alignment between video and textual queries has attracted increasing attention in recent years for MR and HD.

Early models [5,14] typically feed the extracted visual and textual representations into a modality interaction module to produce joint features. However, due to the inherent semantic and structural discrepancy between visual and textual modalities, such direct fusion often leads to suboptimal feature alignment and limited discriminative capability [15].

To address this issue, several recent methods explicitly improve multimodal alignment through architectural design. UVCOM [7] introduces

a multi-aspect contrastive learning strategy to bridge the gap between the local perception required for MR and the global context understanding needed for HD. TR-DETR [3] constructs local and global alignment regulators, where the local regulator encourages the model to distinguish semantically similar but query-irrelevant clips, while the global regulator enforces semantic consistency between video and textual representations in a shared latent space. VideoLights [16] introduces a Feature Refinement and Alignment (FRA) module combined with an alignment loss to refine video-text representations.

Although these methods have significantly advanced multimodal alignment for MR and HD, their alignment mechanisms are still mainly based on contrastive supervision, regulator-based constraints, or attention-derived soft correspondence. Such designs encourage video-text consistency but do not explicitly optimize the matching relationship between each video clip and each textual token. In contrast, our MomentMamba introduces an OT-based local alignment mechanism that formulates clip-token matching as a transport problem. The learned transport matrix provides an explicit fine-grained correspondence between temporally localized visual clips and query tokens, which complements global video-text contrastive alignment and provides better-aligned inputs for subsequent cross-modal interaction and temporal decoding.

2.2. Mamba for efficient sequence modeling

Mamba [9] achieves strong performance on long sequences by significantly reducing computational and memory overhead. This is enabled by its input-dependent state transitions and hardware-aware implementations, which allow for linear-time sequence modeling. VMamba [17] bridges the gap between the sequential nature of one-dimensional selective scanning and the non-sequential structure of two-dimensional visual data by extending selective state space modeling to vision tasks. To further adapt Mamba for video understanding, VideoMamba [18] introduces spatiotemporal scanning strategies and bidirectional state space modeling, enabling efficient long-range temporal modeling with linear complexity, which is particularly suitable for long video sequences. Multi-Scale VMamba [19] further introduces hierarchical scanning strategies to improve representation capacity while maintaining computational efficiency under limited computation budgets. Recent works have also explored Mamba in video retrieval. MUSE [20] introduces a multi-scale Mamba learner for text-video retrieval to model cross-resolution video representations efficiently. MamFusion [21] applies Multi-Mamba and temporal fusion modules to partially relevant video retrieval for better long-sequence video-text interaction. Different from them, our method focuses on joint moment retrieval and highlight detection, and further combines OT-based alignment with collaborative Mamba modeling for precise temporal grounding.

Although these studies demonstrate the potential of Mamba for video-text retrieval, they mainly focus on moment retrieval. In contrast, our work addresses joint moment retrieval and highlight detection, where the model must simultaneously produce accurate temporal boundaries and dense clip-level saliency scores. Moreover, MomentMamba is not a simple replacement of Transformer layers with Mamba blocks. Instead, it combines OT-based local-global alignment, collaborative cross-modal Mamba interaction, and Mamba-based temporal decoding to support fine-grained temporal grounding under a unified MR-HD framework.

3. Method

3.1. Problem definition

The joint task of video Moment Retrieval (MR) and Highlight Detection (HD) aims to identify specific temporal segments and salient clips within a video based on a natural language query. Formally, given an untrimmed video V consisting of L_v sequential clips $[v_1, v_2, \dots, v_{L_v}]$ and

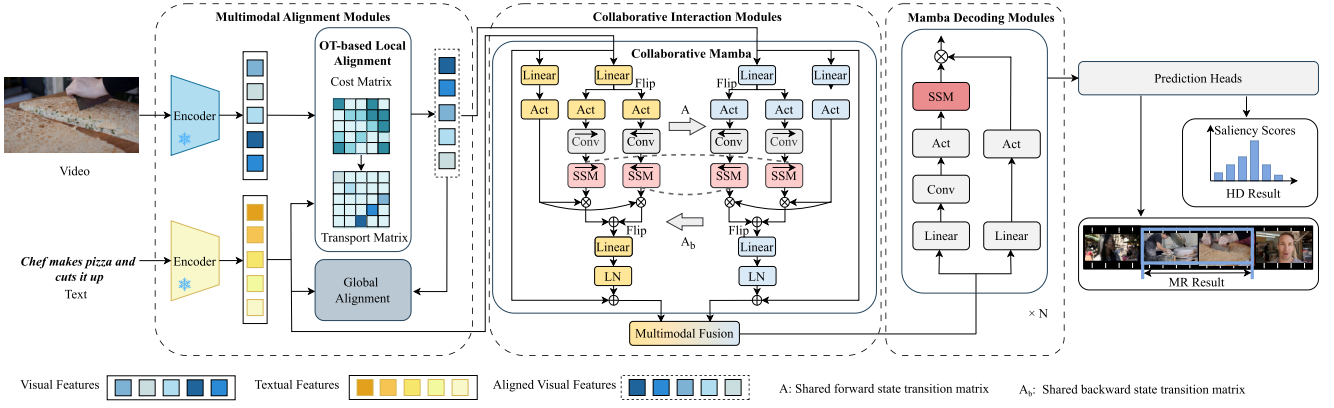


Fig. 2. Overview of MomentMamba. Left: Multimodal alignment modules including OT-based local alignment and global alignment. Middle: Collaborative interaction Mamba with shared state transition matrices. Right: Mamba decoding and task-specific heads for MR and HD.

a textual query T comprising L_t tokens $[t_1, t_2, \dots, t_{L_t}]$, the objective of joint video MR and HD is to simultaneously predict:

1. **Moment Localization:** A temporal tuple (m_c, m_σ) , representing the target moment that is semantically relevant to the given query, where m_c denotes the center timestamp and m_σ represents the duration.
2. **Saliency Scores:** A sequence of scores $P_s = \{s_1, s_2, \dots, s_{L_v}\}$ where each s_i indicates the relevance of the i th video clip to the textual query T .

3.2. Overview of MomentMamba

As illustrated in Fig. 2, the proposed MomentMamba framework is designed for joint MR and HD prediction and consists of three core components: **Multimodal Alignment Module (MAM)**, **Collaborative Interaction Module (CIM)**, and **Mamba Decoding Module (MDM)**. The processing pipeline operates as follows:

1. Modality-specific features are extracted from the raw video V and text T using pretrained video and text encoders, respectively.
2. To bridge the representation gap between visual and textual features, the modality-specific features are aligned, through MAM, by an OT-based local alignment to capture token-clip relevance and a global alignment to enforce video-text consistency between the modalities.
3. The aligned visual and textual features are then fed into CIM, where Collaborative Mamba is employed to enhance multimodal interaction via shared state transition matrices, followed by a multimodal feature fusion to generate unified representations.
4. The fused sequence is decoded by MDM, which consists of stacked selective state space layers [9] designed for efficient long sequence modeling, and then passed to the task-specific prediction heads to generate the final saliency scores P_s and relevant moment coordinates (m_c, m_σ) .

The detailed methodology is presented as follows.

3.3. Feature extraction

3.3.1. Visual features

Following previous work [3,5,6], the input video is segmented into non-overlapping clips with fixed duration. For each clip, we extract visual features using pre-trained ViT-B/32 from CLIP [22] and SlowFast [23]. This produces a sequence of feature embeddings with the dimension d_v : $F_v = [f_{v1}, f_{v2}, \dots, f_{vL_v}] \in \mathbb{R}^{L_v \times d_v}$, where L_v and d_v are the number of video clips and the visual feature dimension, respectively.

3.3.2. Textual features

For a given textual query, we use the CLIP text encoder [22] to produce token-level textual feature embeddings: $F_t = [f_{t1}, f_{t2}, \dots, f_{tL_t}] \in \mathbb{R}^{L_t \times d_t}$, where L_t and d_t represent the number of tokens and the textual feature dimension, respectively.

3.3.3. Auditory features

When training with audio modality, we follow the feature extraction protocol of TR-DETR [3]. Specifically, a pretrained PANN [24] model trained on AudioSet is used to extract clip-level audio features from the original video soundtrack. The audio features are temporally aligned with the same video clip segmentation, resulting in $F_a = [f_{a1}, f_{a2}, \dots, f_{aL_v}] \in \mathbb{R}^{L_v \times d_a}$, where L_v denotes the number of video clips and d_a denotes the audio feature dimension.

The extracted audio features are concatenated with the visual features along the channel dimension: $F_{va} = [F_v; F_a] \in \mathbb{R}^{L_v \times (d_v + d_a)}$. For notation simplicity, in the audio-enhanced setting, F_v in the following formulations denotes the concatenated audio-visual feature F_{va} .

3.4. Multimodal alignment

3.4.1. OT-based local alignment

We adopt an Optimal Transport (OT) strategy [8,25] to achieve precise clip-token alignment. The core idea of OT is to find the most cost-efficient way to transform one probability distribution into another. To apply on our multimodal alignment scenario, OT strategy focuses on aligning visual features to textual features, ensuring both modalities represent the consistent semantic information.

Formally, given the visual features F_v and textual features F_t , we treat each sequence as a discrete distribution and learn a transport matrix, denoted as $M \in \mathbb{R}^{L_v \times L_t}$, which aims to push the embeddings of different modalities closer in the shared space. Beforehand, to perform alignment in a unified embedding space, we project both visual and textual features into a shared d -dimensional embedding space:

$$\hat{F}_v = F_v W_v, \hat{F}_t = F_t W_t, \quad (1)$$

where $W_v \in \mathbb{R}^{d_v \times d}$ and $W_t \in \mathbb{R}^{d_t \times d}$. Accordingly, $\hat{F}_v = [\hat{f}_{v1}, \dots, \hat{f}_{vL_v}] \in \mathbb{R}^{L_v \times d}$ and $\hat{F}_t = [\hat{f}_{t1}, \dots, \hat{f}_{tL_t}] \in \mathbb{R}^{L_t \times d}$. The input dimension of W_v is adjusted according to the visual-side feature dimension, i.e., d_v in the video-only setting and $d_v + d_a$ in the audio-enhanced setting.

In the audio-enhanced setting, F_v in the following formulations denotes the concatenated audio-visual feature F_{va} .

To enforce the alignment of projected features, we define a cost matrix $C \in \mathbb{R}^{L_v \times L_t}$ to present the “cost” required to align each video clip

to each text token by adopting the cosine distance:

$$C_{ij} = 1 - \frac{\langle \hat{f}_{vi}, \hat{f}_{tj} \rangle}{\|\hat{f}_{vi}\|_2 \|\hat{f}_{tj}\|_2}, \quad (2)$$

where $\hat{f}_{vi}$ and $\hat{f}_{tj}$ are the feature vectors of the i th video clip and j th textual token, and $\frac{\langle u, v \rangle}{\|u\| \|v\|}$ represents the cosine similarity. The $\|\cdot\|_2$ denotes the L^2 norm of a vector. Thus, a high similarity corresponds to a low transport cost, indicating highly related features from different modalities are pushed closer in the shared space. Therefore, the OT problem seeks the optimal transport matrix M that minimizes the total transport cost:

$$\min_M \sum_{i=1}^{L_v} \sum_{j=1}^{L_t} M_{ij} C_{ij}. \quad (3)$$

$$\text{s.t.} \quad \sum_{j=1}^{L_t} M_{ij} = \frac{1}{L_v}, \quad \sum_{i=1}^{L_v} M_{ij} = \frac{1}{L_t}, \quad M_{ij} \geq 0. \quad (4)$$

However, strict OT requires the total mass of both distributions to be equal and fully matched, which can be overly restrictive given the noise and redundancy in untrimmed videos, e.g., background clips may not match any text. To address this problem, a relaxed constraint [26] is adopted practically, at the same time improving computational efficiency. We relax the column constraint, normalizing only the rows to ensure that every video clip finds its optimal text token for semantic alignment:

$$\sum_{j=1}^{L_t} M_{ij} = \frac{1}{L_v}, \quad M_{ij} \geq 0. \quad (5)$$

This cost optimization results in an optimal transport matrix M that explicitly aligns video clips to the text tokens of highest relevance, effectively achieving clip-token local alignment between visual and textual features.

Finally, the aligned video features $\hat{F}'_v = [\hat{f}'_{v1}, \hat{f}'_{v2}, \dots, \hat{f}'_{vL_v}] \in \mathbb{R}^{L_v \times d}$ can then be obtained through:

$$\hat{F}'_v = \hat{F}_v + M \hat{F}_t. \quad (6)$$

3.4.2. Global alignment

Although the OT-based local alignment captures clip-token relevance, it is equally critical to enhance video-text consistency, ensuring both modalities share a unified semantic space. To this end, we incorporate a global alignment step [27] to complement the local alignment by aligning the global representations of visual and textual features.

Specifically, a global video representation is computed via the average pooling over the locally aligned visual features $\hat{f}'_{vi}$:

$$G_v = \frac{1}{L_v} \sum_{i=1}^{L_v} \hat{f}'_{vi} \in \mathbb{R}^d. \quad (7)$$

Similarly, the global textual representation is obtained by the average pooling over the textual features $\hat{f}_{tj}$:

$$G_t = \frac{1}{L_t} \sum_{j=1}^{L_t} \hat{f}_{tj} \in \mathbb{R}^d. \quad (8)$$

To enforce alignment in this shared semantic space, we employ a multimodal contrastive loss that, within a batch, pulls the embeddings of originally paired video-text samples closer while pushing apart those of unpaired ones. For a batch of N paired samples, the loss is defined as:

$$\mathcal{L}_{\text{align}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(G_v^i, G_t^i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(G_v^i, G_t^j)/\tau)}, \quad (9)$$

where $\text{sim}(G_v, G_t)$ denotes the cosine similarity $\frac{\langle u, v \rangle}{\|u\| \|v\|}$ between the projected global features, and τ is a temperature hyperparameter. This alignment loss $\mathcal{L}_{\text{align}}$ will be combined with the final prediction loss for joint optimization, stabilizing the visual and textual feature alignment.

3.5. Collaborative interaction

To enhance the interaction between visual and textual modalities, promoting query-relevant information while weakening query-irrelevant information, we introduce the Collaborative Interaction Module (CIM). The core of this module is the Collaborative Mamba, which extends the State Space Models (SSM) to multimodal interaction by sharing state transition matrices between two modalities. This design not only effectively captures multimodal correlations, but also ensures the scalability for long video sequences with a linear computational complexity.

3.5.1. Preliminaries: selective state space model

The foundation of our approach is Selective SSM, which yields an output $\mathbf{y}(t)$, from an input $\mathbf{x}(t)$, through an implicit latent state $\mathbf{h}(t)$. It is formulated as a continuous linear time-invariant dynamical system. Formally, it describes the evolution of hidden states over time as follows:

$$\frac{d}{dt} \mathbf{h}(t) = \mathbf{A} \mathbf{h}(t) + \mathbf{B} \mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C} \mathbf{h}(t), \quad (10)$$

where the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are learnable parameters representing state transitions, input projection, and output mapping, respectively.

To apply this to discrete text and video sequences, the continuous system is discretized using a timescale parameter Δ , leading to the following transformations under the zero-order hold assumption:

$$\bar{\mathbf{A}} = \exp(\Delta \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I})\mathbf{B}, \quad (11)$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are the discretized state transition. This yields the time-discrete SSM as follows:

$$\mathbf{h}_t = \bar{\mathbf{A}} \mathbf{h}_{t-1} + \bar{\mathbf{B}} \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C} \mathbf{h}_t. \quad (12)$$

Therefore, the Selective SSM allows the model to selectively remember or ignore information at each timestep by optimizing the learnable parameters $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$.

3.5.2. Feature interaction and multimodal fusion

To adapt inherently unimodal Mamba model for capturing both intra-modal temporal dependencies and inter-modal interactions, we introduce a Collaborative Mamba which has a bimodal, bidirectional structure, as shown in Fig. 2. This design enables parallel processing of visual and textual embeddings, while enhancing cross-modal interaction via shared state transition matrices between modalities.

To simplify the representation of the formulations, we use X_m , where $m \in \{v, t\}$, to represent the $\hat{F}'_v \in \mathbb{R}^{L_v \times d}$ and $\hat{F}_t \in \mathbb{R}^{L_t \times d}$, the embeddings of visual and textual features after local-global alignment. Each modality stream X_m is passed through a modality-specific Mamba block defined as:

$$\tilde{X}_m = \text{SSM}_m \left(\text{Conv}_m \left(\sigma(\mathbf{W}_1^{(m)} X_m) \right) \right) \odot \sigma(\mathbf{W}_2^{(m)} X_m), \quad (13)$$

where $\sigma(\cdot)$ denotes a non-linear activation and $\text{Conv}_m(\cdot)$ is a depthwise 1D convolution for local token mixing, and $\odot$ denotes element-wise multiplication.

The core innovation here lies in SSM parameters sharing, to model both shared and modality-specific temporal dynamics. Specifically, the state transition matrix, denoted as $\mathbf{A}$, is shared across both the visual and textual streams to enable cross-modal interaction, while the input and output projection matrices $\mathbf{B}_m$ and $\mathbf{C}_m$ remain independent, preserving individual modality characteristics of visual and textual features.

Furthermore, we utilize a bidirectional structure, since MR requires understanding future context. The forward process is formulated as:

$$\mathbf{H}_{m,t}^{(f)} = \mathbf{A} \mathbf{H}_{m,t-1}^{(f)} + \mathbf{B}_m \mathbf{X}_{m,t}, \quad \mathbf{Y}_{m,t}^{(f)} = \mathbf{C}_m \mathbf{H}_{m,t}^{(f)}. \quad (14)$$

Accordingly, the backward process is:

$$\mathbf{H}_{m,t}^{(b)} = \mathbf{A}_b \mathbf{H}_{m,t+1}^{(b)} + \mathbf{B}_m \mathbf{X}_{m,t}, \quad \mathbf{Y}_{m,t}^{(b)} = \mathbf{C}_m \mathbf{H}_{m,t}^{(b)}, \quad (15)$$

where $\mathbf{H}_{m,t}^{(f)}$ and $\mathbf{H}_{m,t}^{(b)}$ are the forward and backward hidden states, respectively. Note that the backward transition matrix $\mathbf{A}_b$ is also shared across

modalities. After passing through the Collaborative Mamba, two sequences of fully-interacted features $\tilde{F}_v \in \mathbb{R}^{L_v \times d}$ and $\tilde{F}_t \in \mathbb{R}^{L_t \times d}$ for visual and textual modalities are obtained, followed by a lightweight cross-attention layer to further generate joint features. Specifically, the interacted visual features are projected as queries $Q_v = \text{Linear}_q(\tilde{F}_v)$, keys and values are from the textual features $K_t = \text{Linear}_k(\tilde{F}_t)$, $V_t = \text{Linear}_v(\tilde{F}_t)$.

$$Y_{\text{fused}} = \text{CrossAttention}(Q_v, K_t, V_t) = \text{Softmax}\left(\frac{Q_v K_t^\top}{\sqrt{d}}\right)V_t, \quad (16)$$

where $Q_v \in \mathbb{R}^{L_v \times d}$, $K_t, V_t \in \mathbb{R}^{L_t \times d}$ and $Y_{\text{fused}} \in \mathbb{R}^{L_v \times d}$ represent the joint features through modalities interaction and fusion.

It is worth noting that this cross-attention layer does not undermine the efficiency advantage of MomentMamba. Unlike Transformer-based temporal decoders that repeatedly perform self-attention over the full video sequence with $O(L_v^2 \cdot d)$ complexity, our cross-attention is applied only once between the visual-side sequence and the much shorter textual query sequence, with complexity $O(L_v \cdot L_t \cdot d)$. Since L_t is usually much smaller than L_v in our tasks, this operation grows approximately linearly with the video length. The dominant long-range temporal modeling is still handled by the Mamba-based decoder with linear complexity.

3.6. Mamba decoding

Traditional Transformer decoders [5] rely on heavy cross-attention layers, which leads to quadratic computational complexity. Following the design of collaborative Mamba, we employ a Mamba decoder based on a stack of selective SSM to explore multimodal correlations for task-specific representations. This decoder, which is efficient for long-range sequence modeling, not only accelerates training but also enhances accuracy.

Formally, the decoder takes fused multimodal features Y_{fused} as input, and processes it through a stack of N decoding layers. Each consists of a depthwise 1D convolution for local context aggregation, followed by a non-linear activation, and a selective SSM. A decoding layer is defined as:

$$Z = \text{SSM}(\sigma(\text{Conv}(Y_{\text{fused}}))), \quad (17)$$

where $Z \in \mathbb{R}^{L_v \times d}$ is the output of decoder containing potential correlations between modalities, $\text{Conv}(\cdot)$ is a 1D depthwise convolution and $\sigma(\cdot)$ denotes a non-linear activation function. For stabilizing deep network training, residual connections are utilized.

3.7. Prediction heads and loss

MomentMamba is trained with objectives for moment retrieval and highlight detection, jointly including the local-global alignment.

3.7.1. Moment retrieval

The MR head is designed to localize the target moment boundaries. The head contains two parallel branches: a regression branch that predicts the normalized center and duration (m_c, m_d) for each potential candidate, and a classification branch that determines whether each predicted span is semantically relevant to the textual query, i.e., foreground (relevant) or background (irrelevant).

To ensure precise localization, we employ a composite loss $\mathcal{L}_{\text{MR}}$ that jointly optimizes temporal boundary localization and moment classification. Specifically, the moment loss $\mathcal{L}_{\text{MR}}$ consists of: 1) an L1 regression term $\mathcal{L}_{\text{L1}}$ minimizing the absolute error between predicted and ground-truth coordinates. 2) a generalized Intersection over Union (IoU) loss $\mathcal{L}_{\text{IoU}}$ to handle scale invariance. 3) and a cross-entropy loss $\mathcal{L}_{\text{CE}}$ for foreground/background classification. Given the ground truth boundaries m , predicted boundaries $\hat{m}$, ground truth class labels y , and predicted class $\hat{y}$, the MR loss is defined as:

$$\mathcal{L}_{\text{MR}} = \lambda_{\text{L1}} \mathcal{L}_{\text{L1}}(m, \hat{m}) + \lambda_{\text{IoU}} \mathcal{L}_{\text{IoU}}(m, \hat{m}) + \lambda_{\text{CE}} \mathcal{L}_{\text{CE}}(y, \hat{y}), \quad (18)$$

where λ_{L1} , λ_{IoU} , and λ_{CE} are hyperparameters that balance the contributions of each term.

3.7.2. Highlight detection

For HD, the output from the Mamba Decoder is passed through a lightweight prediction head comprising two fully connected layers to output a saliency score for each clip.

The HD task is treated as a ranking problem. We combine a margin-based loss $\mathcal{L}_{\text{margin}}$ and a ranking contrastive loss $\mathcal{L}_{\text{rank}}$ to optimize clip-wise saliency scores:

$$\mathcal{L}_{\text{HD}} = \lambda_{\text{margin}} \mathcal{L}_{\text{margin}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}, \quad (19)$$

where λ_{margin} and λ_{rank} are weighting hyperparameters.

The margin-based loss $\mathcal{L}_{\text{margin}}$ enforces the model to assign higher saliency scores to higher-ranked clips than to lower-ranked clips by a margin Δ , which is defined as:

$$\mathcal{L}_{\text{margin}} = \max(0, \Delta + S_{\text{low}} - S_{\text{high}}). \quad (20)$$

The rank-aware contrastive loss $\mathcal{L}_{\text{rank}}$ aims to learn finer-grained saliency levels by contrasting sets of saliency scores at different ranks. Given a maximum rank R , for each rank index $r \in \{0, \dots, R-1\}$, let S_r^+ and S_r^- denote the sets of saliency scores higher and lower than r , respectively. The loss is formulated as:

$$\mathcal{L}_{\text{rank}} = - \sum_{r=0}^{R-1} \log \frac{\sum_{s \in S_r^+} \exp(s/\tau)}{\sum_{s \in (S_r^+ \cup S_r^-)} \exp(s/\tau)}, \quad (21)$$

where τ is a temperature hyperparameter controlling the softness of the distribution.

3.7.3. Total training objective

The MomentMamba framework is trained end-to-end by minimizing a weighted sum of the alignment, MR, and HD losses. The total objective is formulated as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{MR}} \mathcal{L}_{\text{MR}} + \lambda_{\text{HD}} \mathcal{L}_{\text{HD}}, \quad (22)$$

where λ_{align} , λ_{MR} and λ_{HD} are hyperparameters balancing each loss term.

4. Experiment

4.1. Datasets

We validate the effectiveness of the MomentMamba framework on three standard benchmarks, covering both joint and individual MR and HD tasks.

QVHighlights Dataset: As the primary benchmark for joint MR and HD, QVHighlights [5] contains over 10,000 YouTube videos covering a wide range of topics, including daily activities and news events. Importantly, each video is paired with a human-written textual query, and dual annotations: specific temporal start/end timestamps for MR tasks and five-point scale saliency scores for all query-relevant clips for HD tasks.

TVSum Dataset: Designed specifically for HD tasks, the TVSum dataset [32] includes 50 videos of 10 distinct categories, such as news, travel, and documentaries. Though the primary task of the dataset is video summarization, it provides shot-level importance scores obtained via crowd-sourcing, which makes it ideal for evaluating models designed for HD tasks.

Charades-STA Dataset: The Charades-STA dataset [33] serves as a benchmark for MR tasks, built upon the original Charades dataset, but having sentence-level temporal annotations. It consists of 9848 videos of common indoor activities and 16,128 manually written textual queries with sentence-level temporal annotations.

4.2. Evaluation metrics and implementation settings

4.2.1. Evaluation metrics

We use evaluation metrics established in previous works [3,5,6]. For MR performance, we report Recall@1 (R1) at IoU thresholds of 0.5 and 0.7, denoted as **R1@0.5** and **R1@0.7**. Additionally, we compute the

Table 1

Experimental results on the QVHighlights test set. HD represents the results of HD. † represents training with audio modality. Bold letters indicate the best results, while underlined results are suboptimal.

Method	MR					HD	
	R1		mAP			≥ Very Good	
	@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1
XML+ [5]	46.69	33.46	47.89	34.67	34.90	35.38	55.06
MDETR [5]	52.89	33.02	54.82	29.40	30.73	35.69	55.60
QD-DETR [14]	62.40	44.98	62.62	39.88	39.86	38.64	62.40
UniVTG [28]	58.86	40.86	57.60	35.59	35.47	38.20	60.96
MH-DETR [29]	60.05	42.48	60.75	38.13	38.38	38.22	60.51
TR-DETR [3]	64.66	48.96	63.98	43.73	42.62	39.91	63.42
CG-DETR [30]	65.43	48.38	64.51	42.77	42.86	40.33	<u>66.21</u>
UVCOM [7]	63.55	47.47	63.37	42.67	43.18	39.74	64.20
MS-DETR [31]	64.72	48.77	66.41	<u>44.91</u>	44.89	<u>40.45</u>	65.95
Diff-DETR [4]	<u>66.23</u>	51.12	65.25	<u>43.87</u>	44.03	<u>40.11</u>	65.21
MomentMamba (Ours)	67.12 ± 0.65	48.92 ± 0.41	<u>66.06 ± 0.53</u>	45.01 ± 0.31	<u>44.38 ± 0.45</u>	40.62 ± 0.70	66.39 ± 0.68
UMT† [6]	56.23	41.18	53.38	37.01	36.12	38.18	59.99
QD-DETR† [14]	63.06	45.10	63.04	40.10	40.19	39.04	62.87
UVCOM† [7]	63.81	48.70	64.47	<u>44.01</u>	43.27	39.79	64.79
TR-DETR† [3]	65.05	47.67	64.87	42.98	43.10	39.90	63.88
Diff-DETR† [4]	<u>66.42</u>	51.74	65.73	43.91	<u>44.14</u>	40.23	65.34
MomentMamba† (Ours)	67.32 ± 0.56	<u>50.07 ± 0.36</u>	65.89 ± 0.47	44.45 ± 0.40	44.39 ± 0.50	41.54 ± 0.51	66.27 ± 0.59

mean Average Precision (mAP) at IoU thresholds of 0.5 and 0.75, along with the average mAP calculated over a range of 10 uniformly distributed IoU thresholds {0.5, 0.55, ..., 0.95}. Performance of HD task is evaluated using mAP and **HIT@1 (top-1 hit accuracy)**, measuring how well the predicted saliency scores align with human annotations.

For QVHighlights dataset, we report all the mentioned metrics for both MR and HD evaluation. While we assess temporal localization accuracy using R1@0.5 and R1@0.7 on Charades-STA dataset for the MR task only, and HD performance on TVSum dataset using the mAP computed over the top-5 predictions.

4.2.2. Implementation settings

We implement MomentMamba using PyTorch. All experiments are conducted on NVIDIA V100 GPUs with Intel Xeon Gold 5320 CPUs. To ensure the reliability of our results, we conducted five independent runs with different random seeds under the same configuration to confirm the stability of our method. On QVHighlights, the model is trained for 200 epochs with a batch size of 32 and a learning rate of $1e-4$. While for a smaller dataset TVSum, the training runs for 2000 epochs with a batch size of 4 and a learning rate of $1e-3$. On Charades-STA, we train it for 100 epochs with a batch size of 8 and a learning rate of $1e-4$. We use a weight decay of 1×10^{-4} and apply gradient clipping with a maximum norm of 0.1.

For feature projection and multimodal interaction, both modalities are mapped into a shared embedding space of dimension $d = 256$. In the Mamba-based decoding stage, each Mamba block uses a state dimension of 16, a convolution kernel size of 4, and an expansion factor of 2. The decoder contains 2 layers by default, and dropout is set to 0.1 throughout the model. For OT-based local alignment in MAM, we compute the cosine-distance cost matrix between normalized video and text features for each sample individually. The transport plan M is obtained using Sinkhorn optimization with entropic regularization coefficient $\epsilon = 0.1$. For moment retrieval, the loss weights are set to $\lambda_{L1} = 10$, $\lambda_{gIoU} = 1$, and $\lambda_{CE} = 4$. For highlight detection, we set $\lambda_{margin} = 1$ and $\lambda_{rank} = 1$, with margin $\Delta = 0.2$ and temperature $\tau = 0.5$. For the overall objective, we set $\lambda_{align} = 1$, $\lambda_{MR} = 1$, and $\lambda_{HD} = 1$.

4.3. Comparison with state-of-the-art models

We compare the proposed framework MomentMamba against existing state-of-the-art methods on three benchmarks, demonstrating that our framework achieves SOTA performance for both joint and individual MR and HD tasks.

4.3.1. QVHighlights

As summarized in Table 1, we evaluate MomentMamba on the QVHighlights benchmark under both video-only and audio-enhanced settings.

Video-Only: Using only visual features, MomentMamba obtains remarkable accuracy, recording 67.12% for R1@0.5, 48.92% for R1@0.7, 66.06% for mAP@0.5, 45.01% for mAP@0.75, and 44.38% for average mAP. Notably, compared to the recent strong baseline Diff-DETR, our method improves R1@0.5 by 0.89% and achieves the highest mAP@0.75, indicating the superior precision in tight boundary localization. For HD, our method achieves the best results with 40.62% mAP and 66.39% HIT@1, outperforming all competing methods. This result reflects the effectiveness of MomentMamba in modeling long-range temporal dependencies and filtering redundant contextual information.

Audio-Enhanced: When audio modality (†) is integrated, our framework extends its capabilities, achieving 67.32% for R1@0.5, 50.07% for R1@0.7, 65.89% for mAP@0.5, and 41.54% for HD mAP. This leading performance further proves that our Collaborative Mamba architecture can effectively integrate multimodal inputs.

4.3.2. TVSum

We further evaluate the HD capabilities of MomentMamba on the TVSum dataset, as shown in Table 2.

Video-Only: MomentMamba achieves the highest average mAP of 88.8%. Remarkably, MomentMamba obtains the best performance in the majority of categories, including Vehicle Tire (VT), Vehicle Unstuck (VU), PaRade (PR), Flash Mob (FM), Bike Tricks (BT), and Dog Show (DS). These results indicate that the temporal modeling of the proposed framework generalizes well across diverse video content.

Audio-Enhanced: When audio features (†) are incorporated, our model maintains its leadership with the highest average score of 88.9%, outperforming TR-DETR and QD-DETR. Notably, our method obtains the best or second-best performance in most individual categories, further confirming the effectiveness of our model in multimodal scenarios.

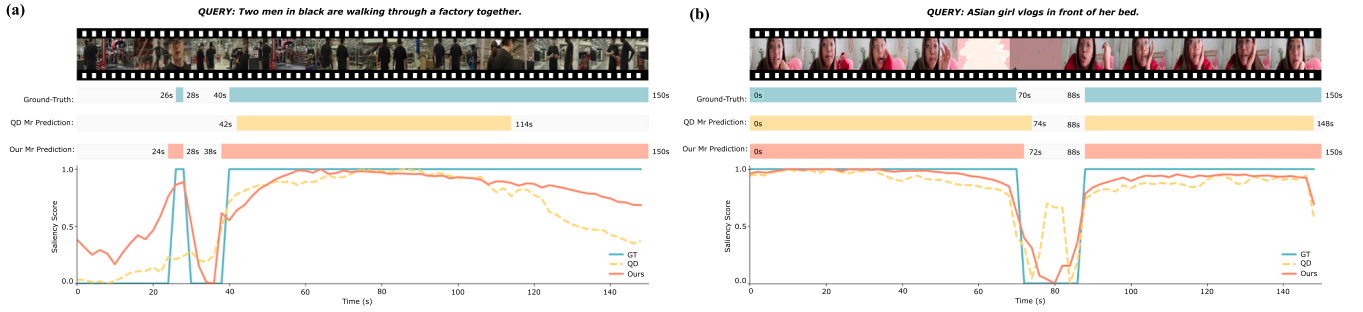
4.3.3. Charades-STA

As shown in Table 3, we evaluate our method for MR performance on the Charades-STA test set using SlowFast + CLIP (SF+C) features. MomentMamba reaches 60.64% for R1@0.5 and 40.36% for R1@0.7, surpassing all existing state-of-the-art methods under the same feature setting. Specifically, compared to the previous best result by MS-DETR [31], our method surpasses it by a significant improvement of

Table 2

Experimental results of HD performance on the TVSum val set. † represents training with audio modality.

Method	VT	VU	GA	MS	PK	PR	FM	BK	BT	DS	Avg
sLSTM [34]	41.1	46.2	46.3	47.7	44.8	46.1	45.2	40.6	47.1	45.5	45.1
SG [35]	42.3	47.2	47.5	48.9	45.6	47.3	46.4	41.7	48.3	46.6	46.2
LIM-S [36]	55.9	42.9	61.2	54.0	60.3	47.5	43.2	66.3	69.1	62.6	56.3
Trailer [37]	61.3	54.6	65.7	60.8	59.1	70.1	58.2	64.7	65.6	68.1	62.8
QD-DETR [14]	88.2	87.4	85.6	85.0	85.8	86.9	76.4	91.3	89.2	73.7	85.0
UniVTG [28]	83.9	85.1	89.0	80.1	84.6	87.0	70.9	91.7	73.5	69.3	81.0
TR-DETR [3]	90.3	93.0	94.3	85.1	88.0	88.6	80.4	91.3	89.5	81.6	88.1
MS-DETR [31]	89.8	93.3	94.8	88.7	88.5	89.0	80.5	94.0	88.7	76.9	88.4
Diff-DETR [4]	90.2	88.4	92.8	87.2	86.5	87.9	75.6	93.2	88.2	75.8	86.6
MomentMamba (Ours)	90.5	93.5	94.6	87.6	88.4	89.8	80.7	93.6	89.6	79.7	88.8
MINI-Net† [38]	80.6	68.3	78.2	81.8	78.1	65.8	75.8	75.0	80.2	65.5	73.2
TCG† [39]	85.0	71.4	81.9	78.6	80.2	75.5	71.6	77.3	78.6	68.1	76.8
UMT† [6]	87.5	81.5	88.2	78.8	81.4	87.0	76.0	86.9	84.4	79.6	83.1
QD-DETR† [14]	87.6	91.7	90.2	88.3	84.1	88.3	78.7	91.2	87.8	77.7	86.6
TR-DETR† [3]	90.6	92.4	91.7	81.3	86.9	85.5	79.8	93.4	88.3	81.0	87.1
Diff-DETR† [4]	90.7	89.2	93.1	86.4	87.5	86.8	76.7	93.6	88.0	78.3	87.0
MomentMamba† (Ours)	90.8	92.4	94.3	88.0	88.3	89.8	81.0	93.8	89.3	81.2	88.9

**Fig. 3.** Qualitative results of MomentMamba compared to baseline QD-DETR on QVHighlights val set.**Table 3**

Experimental results of MR performance on the Charades-STA test set. † represents training with audio modality. SF + C denotes video features extracted using SlowFast and CLIP.

Method	Feat	R1@0.5	R1@0.7
SAP [40]	VGG	27.42	13.36
TripNet [41]	VGG	36.61	14.50
FVMR [27]	VGG	42.36	24.14
UMT† [6]	VGG	48.31	29.25
QD-DETR [14]	VGG	52.77	31.13
CTRL [33]	C3D	23.63	8.89
ACL [42]	C3D	30.48	12.20
MAN [1]	C3D	46.53	22.72
QD-DETR [14]	SF + C	57.31	32.55
UniVTG [28]	SF + C	58.01	35.65
TR-DETR [3]	SF + C	57.61	33.52
CG-DETR [30]	SF + C	58.44	36.34
UVCOM [7]	SF + C	59.25	36.64
MS-DETR [31]	SF + C	59.62	36.48
MomentMamba(Ours)	SF + C	60.64 ± 0.35	40.36 ± 0.42

1.02% for R1@0.5 and 3.88% for R1@0.7. The larger gain under stricter IoU thresholds highlights the advantage of the proposed OT-based local-global alignment in producing more accurate temporal boundaries. Furthermore, our method consistently outperforms representative DETR-based models, such as QD-DETR [14], UniVTG [28], and TR-DETR [3], demonstrating its superior temporal localization capability and generalization ability across datasets.

4.4. Ablation study

To validate the effectiveness of our proposed modules, we conduct ablation studies on the QVHighlights validation set. We systematically

analyze the contributions of three core components in the proposed MomentMamba framework: (1) the **Multimodal Alignment Module (MAM)**, (2) the **Collaborative Interaction Module (CIM)**, and (3) the **Mamba Decoding Module (MDM)**. The results are summarized in Table 4.

Row (a) represents the baseline model QD-DETR [14], a standard Transformer-based architecture without our proposed innovations. Row (b)–(d) evaluate the individual impact of adding each module to the baseline, row (e)–(g) evaluate the effect of combining two modules (equivalently, removing one) to assess inter-module dependencies, and row (h) represents the full MomentMamba framework.

Effect of individual component. Compared row (a) and (b), incorporating the MAM in the baseline model significantly improves performance, increasing the average mAP from 40.60% to 43.53% and HIT@1 from 61.03% to 62.27%. This confirms the effectiveness of early-stage OT-based alignment for reducing the semantic gap between modalities. In row (c), the introduction of the CIM enhances temporal modeling and improves both R1@0.7 and mAP@0.75, resulting in an average mAP of 41.88%. This validates that our Collaborative Mamba module successfully captures cross-modal dependencies more effectively than standard attention mechanisms. Row (d), which replaces the transformer decoder with the proposed Mamba decoder, also shows the improvements in both MR and HD metrics, validating its role in long-range sequence modeling.

Analysis of synergistic components. In row (e)–(g), the synergy between modules is further analyzed. Row (e) shows that excluding the CIM results in a performance drop compared to the full model in row (h). This indicates that the interaction is crucial for deeply fusing the modalities. Omitting the alignment module shown in row (f), leads to lower retrieval precision, particularly in mAP@0.5 and HD metrics. It reflects that without the clip-token local alignment, the downstream modules struggle to process unaligned features, introducing noise to the

Table 4

Ablation study of each module in MomentMamba on the QVHighlights validation split.

Setting	MAM	CIM	MDM	MR					HD	
				R1		mAP			$\geq$ Very Good	
				@0.5	@0.7	@0.5	@0.75	Avg.	mAP	HIT@1
(a)				61.55	45.16	62.00	41.05	40.60	39.04	61.03
(b)	✓			63.84	47.69	63.77	43.03	43.53	38.95	62.27
(c)		✓		63.90	47.91	63.45	42.18	41.88	39.35	61.84
(d)			✓	62.42	46.12	62.80	41.92	41.30	39.28	61.47
(e)	✓		✓	64.75	48.30	64.15	43.37	43.02	39.85	64.03
(f)		✓	✓	64.90	48.56	64.48	43.71	43.40	40.06	63.42
(g)	✓	✓		65.06	48.93	65.58	43.22	42.85	39.74	64.71
(h)	✓	✓	✓	66.83	50.96	66.30	44.56	44.67	41.86	66.69

Table 5

Efficiency comparison between transformer-based and Mamba-based decoders.

Decoder	Params (M)	FLOPs (M)	Complexity	Training Time
Transformer-based (Baseline)	9.24	286.38	$\mathcal{O}(L^2 \cdot d)$	1.00×
Mamba-based (Ours)	8.58	260.54	$\mathcal{O}(L \cdot d)$	0.91×

prediction phase. Row (g) excludes the Mamba Decoder and results in reduced performance in both R1 and mAP metrics, underscoring its role in capturing long-term dependencies. The complete model in setting (h) achieves the best overall performance, showing the complementary effectiveness of all three modules.

4.5. Efficiency analysis

To evaluate the computational efficiency of the proposed framework, we conduct quantitative comparisons on model efficiency against the Transformer-based baseline, QD-DETR. To ensure a fair comparison, we compare only the proposed Mamba-based decoder with a Transformer-based decoder from QD-DETR, while keeping the identical settings of the feature encoders, alignment modules, hidden dimensions, and training configurations. Training time is measured as the average wall-clock time per iteration and normalized to the QD-DETR baseline. As summarized in Table 5, the proposed Mamba-based decoder achieves consistently lower computational cost and improved training efficiency. In particular, it reduces both parameter count and FLOPs while decreasing the theoretical complexity from quadratic $\mathcal{O}(L^2 \cdot d)$ to linear $\mathcal{O}(L \cdot d)$. In practice, we observe a 9% reduction in average training time per iteration compared to the baseline model with a Transformer-based decoder. These results confirm that MomentMamba not only achieves state-of-the-art accuracy, but does so with superior scalability, making it a highly practical solution for real-world video understanding tasks.

4.6. Sensitivity analysis of Sinkhorn regularization

To further investigate the influence of the Sinkhorn regularization coefficient ϵ in OT-based local alignment, we conduct a sensitivity analysis on the QVHighlights validation split. The coefficient ϵ controls the smoothness of the transport plan. A smaller value encourages sharper clip-token matching but may be sensitive to noisy or irrelevant clips, while a larger value produces smoother transport distributions but may weaken fine-grained local correspondence.

As shown in Table 6, $\epsilon = 0.1$ achieves the best or near-best performance across major MR and HD metrics. This indicates that $\epsilon = 0.1$ provides a good trade-off between alignment sharpness and optimization stability. Therefore, we adopt $\epsilon = 0.1$ as the default setting in all experiments.

Table 6Sensitivity analysis of the Sinkhorn regularization coefficient ϵ on the QVHighlights validation split.

ϵ	MR			HD	
	R1@0.5	R1@0.7	Avg. mAP	mAP	HIT@1
0.01	65.92	50.11	43.98	41.32	65.84
0.05	66.41	50.52	44.31	41.61	66.20
0.10	66.83	50.96	44.67	41.86	66.69
0.20	66.25	50.34	44.18	41.47	66.08
0.50	66.07	49.73	43.76	41.05	65.41

4.7. Qualitative analysis and visualization

To provide intuitive insights into the effectiveness of MomentMamba, we visualize qualitative results on the QVHighlights dataset in Fig. 3, comparing predictions against the baseline QD-DETR.

As shown in the MR rows, MomentMamba predicts temporal boundaries that are significantly tighter and more aligned with the ground truth, effectively filtering irrelevant context to pinpoint the exact semantic start and end times. In the HD curves, MomentMamba produces smoother and more continuous saliency scores that closely follow the ground-truth distribution. This visual example demonstrates that MomentMamba successfully predicts precise moment boundaries.

5. Conclusion

In this paper, we present MomentMamba, an efficient framework for joint MR and HD. MomentMamba integrates OT-based multimodal alignment modules for both clip-token and video-text semantic modeling, a bidirectional Collaborative Mamba module for cross-modal interaction, and a Mamba decoder for efficient long-sequence decoding. By adopting selective state space modeling, the proposed framework not only achieves precise MR and HD prediction, but also reduces computational complexity from quadratic to linear.

Extensive experiments on public benchmarks demonstrate the effectiveness and efficiency of MomentMamba, achieving strong performance while significantly reducing computational cost compared to Transformer-based baselines.

The linear-complexity temporal modeling of the Mamba decoder improves the scalability of MomentMamba for long video sequences. Nevertheless, extremely long videos, such as videos longer than one hour, may still introduce practical challenges, including increased feature extraction cost, memory consumption, and the accumulation of query-irrelevant temporal noise. In future work, we will explore hierarchical temporal segmentation and memory-efficient streaming inference to further improve the stability and scalability of MomentMamba under extremely long video scenarios.

Moreover, the current audio-enhanced setting suggests that MomentMamba can be naturally extended to additional modalities. Specifically,

auxiliary modalities can be incorporated by temporally aligning them with video clips, extracting modality-specific representations, and projecting them into the shared feature space for lightweight fusion. Therefore, future extensions may include speech transcripts, OCR information, or other auxiliary signals to further enhance fine-grained temporal grounding.

CRedit authorship contribution statement

Yujia Yu: Writing – original draft, Visualization, Validation, Data curation; **Hongjian Shi:** Writing – review & editing, Supervision, Resources, Project administration; **Bingqing Qu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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